

ROHAN N. DHAMDHERE

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[LinkedIn](#) | [Google Scholar](#) | [My Website](#)

SUMMARY

Biomedical engineer and PhD Candidate (Georgia Tech & Emory) with 8+ years building and validating AI models on real-world clinical data across retinal imaging, ECG, echocardiography, and CT. Proven track record translating multimodal patient data into actionable risk tools that outperform established clinical models, with **25+ peer-reviewed publications (100+ citations)** and experience spanning medical centers, a national laboratory, and industry R&D.

RESEARCH EXPERIENCE

Ph.D. Candidate | Georgia Tech & Emory University Aug 2022 – Aug 2026

- ▶ **CARE-CKD:** *Focus:* Cardiovascular risk in CKD. *Novelty:* Retinal vascular biomarkers, Outperforms AHA-PREVENT calculator
Retinal vascular AI risk model for CKD patients outperforming PREVENT calculator (C-Index: 0.70 vs. 0.65); supported by 2025 Georgia Research Alliance translational award for clinical deployment [[Kidney360, 2025](#)]
- ▶ **STAR-Echo:** *Focus:* Ultrasound based Cardiac risk. *Novelty:* Vision Transformer + Radiomics fusion, Interpretable ultrasound AI
Interpretable cardiac ultrasound biomarker (AUC: 0.71) fusing Vision Transformer and Radiomics features for echocardiography-based cardiovascular risk stratification [[MICCAI, 2023](#)]
- ▶ **Heart Transplant ECG:** *Focus:* Non-invasive surveillance, Biopsy reduction *Novelty:* ECG foundation model, longitudinal analysis
Applied ECG foundation model embeddings to longitudinal 12-lead ECGs for non-invasive heart transplant rejection surveillance; achieved 15.1% biopsy reduction (vs. 6.9% clinical baseline) at 95.9% sensitivity and 3x elevated 90-day rejection hazard ratio.

Research Assistant | Center for Computational Imaging & Personalized Diagnostics Aug 2020 – Jul 2022

- ▶ Developed a probabilistic coronary atlas for cardiac CT enabling automated, patient-specific coronary artery disease characterization and territory-level risk mapping [[AHA Scientific Sessions, 2021](#)]
- ▶ Designed a multi-organ CT segmentation pipeline (liver, spleen) for opportunistic hepatic steatosis screening; validated across multi-site cohorts in large-scale cardiovascular outcomes studies [[Lancet eBioMedicine, 2022 & 2025](#)]

Research Assistant | Machine Intelligence Lab, RIT Dec 2016 – Nov 2018

- ▶ Built Graph CNN, object detection, and classification pipelines for traffic sign and stamp feature identification; collaborated with VisualDx Inc. to improve clinical skin lesion diagnosis (+3% accuracy)

INDUSTRY EXPERIENCE

R&D Data Science Intern | Alcon | Johns Creek, GA May – Aug 2025

- ▶ Built a vision-language model prototype fusing corneal tomography, clinical notes, and patient lifestyle data to recommend personalized intraocular lenses for cataract surgery (n=450 patients; hallucination score: 0.13, low is better)

Research Associate | Oak Ridge National Laboratory Dec 2018 – Mar 2020

- ▶ Developed ML/CV algorithms for building and grave detection in satellite imagery via semi-supervised learning (20% labeled data); deployed as an automated end-to-end large-scale counting pipeline [[IEEE GRSL, 2020](#)]

EDUCATION

Ph.D. Biomedical Engineering (Candidate) | Georgia Tech & Emory University 2022 – 2026

- ▶ GPA: 4.0 | Advisor: Dr. Anant Madabhushi | Focus: ML with low-cost clinical modalities (retinal imaging, ECG, echo, CT) for cardiovascular risk prediction

M.S. Computer Engineering | Rochester Institute of Technology 2016 – 2018

- ▶ GPA: 3.90 | Graduate Scholarship Recipient | Thesis: Meta Learning for Graph Neural Networks

TECHNICAL SKILLS

Languages: Python, MATLAB, C++, Agentic AI workflows, Claude Code

ML / Deep Learning: Pytorch, Transformers (ViT, LLM/VLM), CNNs, Graph Neural Networks, Survival Analysis, Foundation Models

Medical Imaging & Analysis: DICOM, NiftI, Echocardiography, CT, Retinal Imaging, Radiomics, Longitudinal Modeling

OTHER SELECT PUBLICATIONS

1. Modanwal G, Dhamdhere R, et al. QuLF-CT: Liver fat quantification on cardiac CT. [[JACC Advances, 2024](#)]
2. Dhamdhere R, et al. Semi-supervised building & grave detection in satellite imagery. [[IEEE GRSL, 2020](#)]
3. Dominguez M, Dhamdhere R, et al. General-Purpose Deep Point Cloud Feature Extractor. [[IEEE WACV, 2018](#)]

TEACHING & LEADERSHIP

- ▶ Co-President, Graduate Student Advisory Board, BME, Georgia Tech & Emory (2023–present); TA: Bioengineering Statistics (Georgia Tech), Deep Learning (RIT)